



DESIGN AND DEVELOPMENT OF A LARYNGOSCOPE

DEPARTMENT OF BIOMEDICAL ENGINEERING

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PRODUCT DESIGN AND PROTOTYPING

Submitted By-

Sonali Chowdhury-BM25MTECH11004

Ankit Chaudhary-BM25MTECH14005

Ankita Das-BM25MTECH14010

DESIGN AND DEVELOPMENT OF A LARYNGOSCOPE

1. Introduction

A laryngoscope is an indispensable medical instrument for the viewing of the larynx and vocal cords, which are key human airway structures. The tool finds frequent application in anesthesiology, otolaryngology, critical medicine, and emergency medicine. By offering an excellent view of the laryngeal inlet, the laryngoscope provides for numerous diagnostic and therapeutic interventions, with its most important use being endotracheal intubation. This operation is crucial for the establishment of an open airway, effective ventilation, and allowing the safe dosing of anesthesia. Laryngoscope invention transformed airway management with its development subsequently becoming an ongoing process for the satisfaction of clinical needs for safety, efficacy, and versatility.

The clinical relevance of laryngoscopes also goes beyond ordinary airway management. They are an indispensable part in the management for conditions involving trauma, respiratory distress, or surgical procedures that need mechanical ventilation. Aside from stabilizing the airway, laryngoscopes also help in the diagnosis for laryngeal pathologies like tumors, infection, or vocal cord paralysis. Laryngoscopes also give therapeutic assistance by aiding procedures for foreign body removal and biopsy sampling. Use has also lessened morbidity and mortality due to airway obstruction, so they are considered one of the foundations of current medicine.

Over time, advancements in design and technology have led to the development of three major types of laryngoscopes: direct, indirect, and video laryngoscopes. The direct type uses a blade and light source to visualize the larynx through a straight line of sight. Indirect laryngoscopes, on the other hand, employ mirrors or optical mechanisms to project the view of the vocal cords without requiring direct alignment. The most recent innovation is the video laryngoscope, which incorporates a miniature camera and display monitor, offering an enlarged and clearer image of the airway. This advancement is especially valuable for managing difficult airways and has also become an effective tool in teaching and training settings. The continued evolution of this device reflects the growing emphasis on patient safety, precision, and clinical versatility.

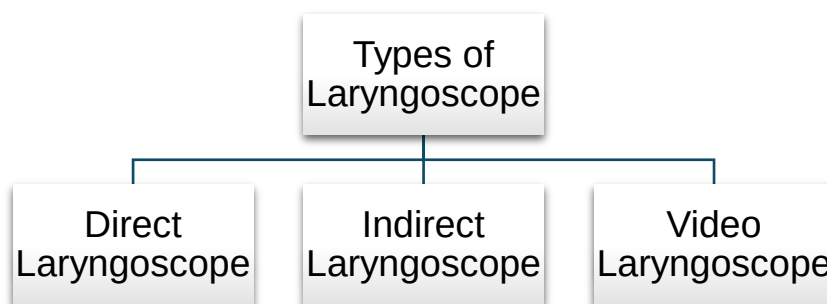


Figure: Types of Laryngoscopes

2. Need and Purpose of the Device

The fundamental function of a laryngoscope is to provide for safe and trustworthy observation of the larynx. This function is the basis for many lifesaving as well as diagnostic uses in medicine. One of the most vital uses is for the purpose of endotracheal intubation, where the trachea is intubated with the help of tube placement to establish the airway and provide for mechanical ventilation. This intervention is essential in both scheduled surgical procedures as well as emergencies, including traumatic or cardiac arrest. Without the capability for observation of the laryngoscope, intubation would be unsafe and very difficult, resulting in major complications involving hypoxia as well as death.

In addition to airway management, the laryngoscope is also indispensable in the diagnosis of laryngeal diseases. Through it, clinicians are able to visualize structural and functional diseases of the vocal cords and nearby tissues. Tumors, infection, inflammation, and vocal cord paralysis can all be correctly diagnosed with its assistance. In addition, the tool comes in handy during therapeutic procedures, such as the removal of foreign bodies that are impaled in the airway as well as taking biopsy specimens for additional pathological studies. Such uses illustrate the multipurpose nature of the laryngoscope in facilitating both relief procedures and long-term management of airway diseases.

The newer generations of the laryngoscope, especially the video laryngoscopes, have also advanced its function. With high-definition visualisation and minimal requirement for the direct line of sight, they not only enhance patient outcomes in challenging airway conditions but also contribute to the process of medical education and training. Educators can monitor the airway on the monitor screen simultaneously by the performing person, facilitating better learning with no compromise on the safety of the patient. Hence, the requirement for the laryngoscope is not only for its clinical application but also for its role in the process of training and education in the field of medicine.

In other words, the device ensures patient safety, high procedural efficacy, and broad clinical usability, making it an indispensable part in modern medicine.

3. Working Principle and Mechanism of Action

The working mechanism of the laryngoscope relies on its power to light up and reveal the laryngeal structures for direct or indirect viewing. In structural composition, the instrument has three major parts: the handle, where the power supply (battery) is located; the blade, which the clinician introduces through the patient's mouth to elevate the tongue and the epiglottis; and the light source, by which the airway is illuminated. Through the working combination of these parts, the laryngeal inlet and vocal cords are rendered visible to the clinician.

The clinician holds the handle during use and inserts the blade into the oral cavity gently. The straight or curved blade is then moved to displace the tongue and raise the epiglottis, opening the larynx. Built-in light sources, formerly always bulbs but increasingly often fiber-optic or light-emitting diode systems, give excellent shadow-free light for easy observation of the

airway. In current video laryngoscopes, a tiny camera at the tip of the blade relays real-time images to an observation monitor for even better visualization.

The mechanism of action varies somewhat for the various classes of the laryngoscopes:

- With direct laryngoscope, the doctor directly looks through the oral opening in order to visualize the larynx.
- Optical prisms or mirrors are used in the indirect laryngoscopy for reflecting the shape of the vocal cords.
- Video laryngoscopy uses a digital lens to send greatly enlarged images through to a monitor so an unbroken line of sight is not required.

Regardless of the type, the principles are the same to provide an unobstructed safe view of the larynx to aid intubation, diagnosis, and therapeutic interventions. With the combination of the ergonomic design, high-quality optics, and robust light sources, the laryngoscope is one of the most versatile tools in airway management.

4. System and Subsystem Level

Laryngoscope works as an airway visualization system and also integrates parts like light, vision, and a modern structure. It ensures clear vision of the larynx. Also focuses on safety and ergonomics. Reliable in Normal checkup or in emergency use.

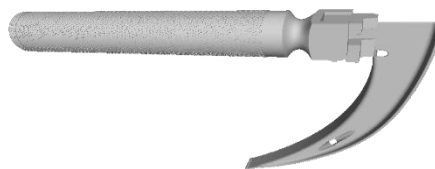


Figure: Laryngoscope

In subsystem level we have:

1. Blade: It lifts the tongue and epiglottis for airway visibility and also gives direct access to laryngeal opening.
2. Light Source: Illuminates the airway region, uses LED or fibre-optic for bright and shadow free light.
3. Video System: provides Direct view from lens or indirect via camera or monitor. Ensures clear visualization of vocal cords.



Figure 4.2: MACINTOSH BLADE

4.Handle: It provide house to the batteries and power supply to the light source and camera. Gives firm grip for easy maneuvering.

5. Power Supply: supplies energy to light and camera usually have rechargeable batteries and an external source.

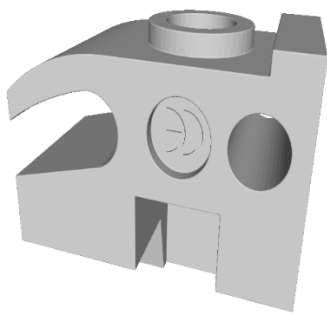


Figure: Base part



Figure: Handle

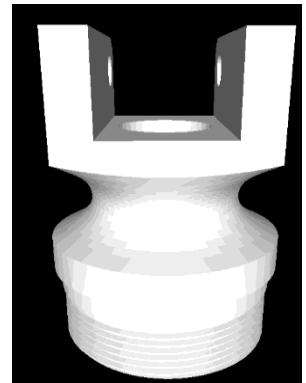
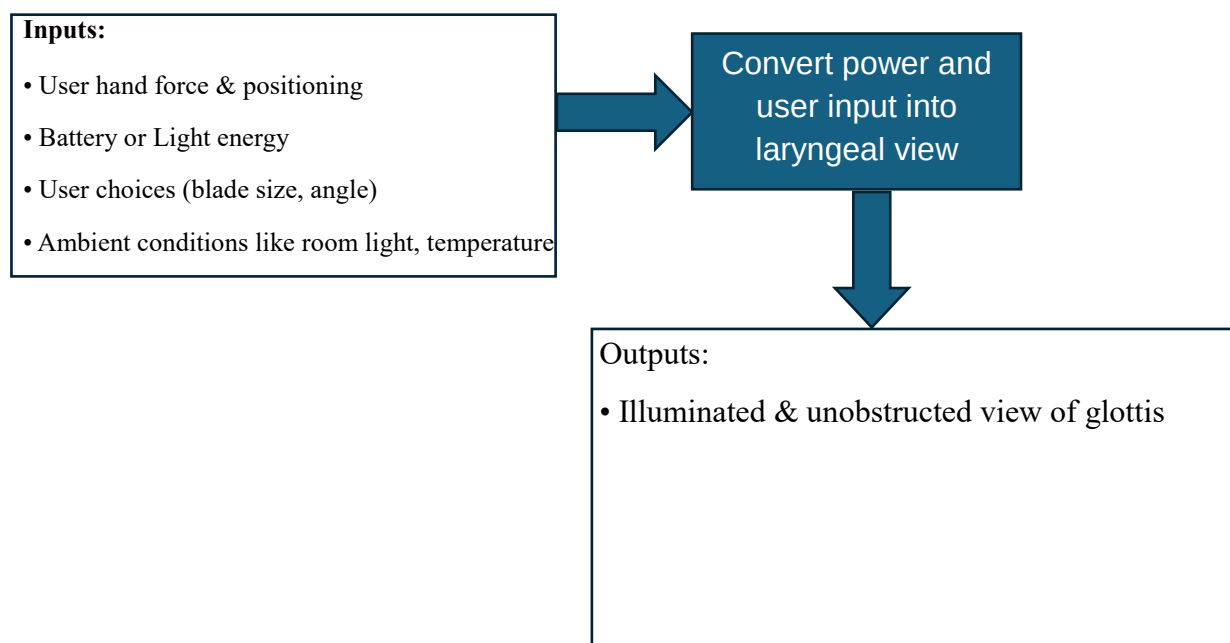


Figure: Holder

5. Reverse Engineering of the device with system and sub-system level design

1. Investigation, Prediction and Hypothesis:

I. Black-Box model for laryngoscope



- Mechanical displacement of tongue / epiglottis
- Heat & stray light
- User feedback like visualization and status

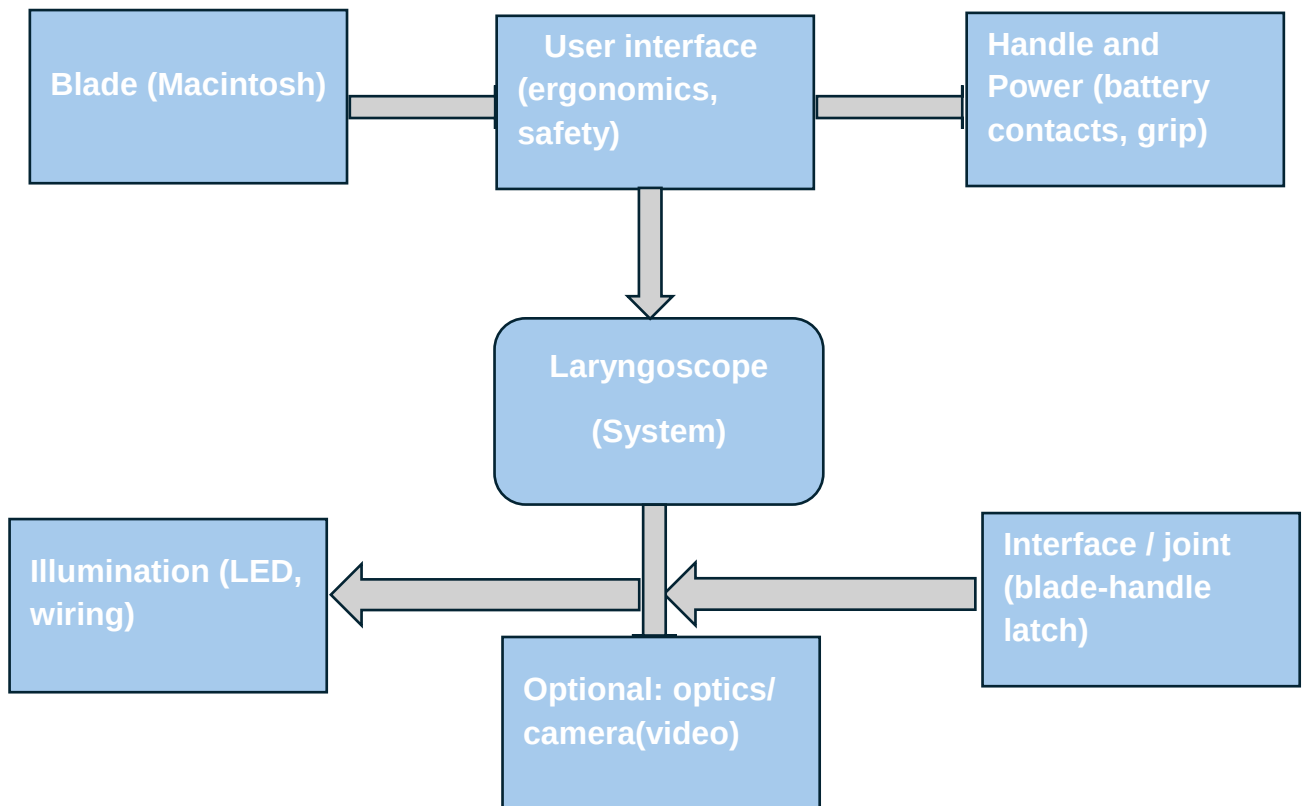


Figure: Subsystem and interaction

Reverse Engineering Steps

1. Preparation

Objective: To gather all necessary background information and tools before starting.

Steps:

- **Obtain the laryngoscope** (traditional or video type – ideally a non-functioning or old model for safe disassembly).
- **Collect manuals and datasheets:** Manufacturer manuals, design specifications, and user guides help understand how the device functions.
- **Research standards:** Look into ISO or ANSI standards related to laryngoscope size, material, and safety.
- **Prepare tools:** Screwdrivers, calipers, multimeter (for electronics), and digital camera.

2. Disassembly

Objective: To carefully take apart the device without damaging parts.

Steps:

- **Separate the handle and blade.**
- If it's a **video laryngoscope**, detach the **camera module**, wires, and display (if any).
- Take note of how the **light source** is connected (usually LED or halogen bulb).
- **Label and store** parts for easy reassembly.

Caution: Some components may be delicate (fiber optics or camera wires).

3. Documentation

Objective: To record all physical and technical details of the components.

Steps:

- **Photograph** each part from multiple angles.
- Use a **vernier caliper or micrometer** to measure:
 - Blade length and curvature.
 - Handle diameter and length.
 - Location and size of screws, power terminals, etc.
- Record part **numbers, materials, textures, and assembly methods** (snap-fit, screw, soldered).

4. Analysis

Objective: To understand the working and functional role of each component.

Key Areas:

a) Material Analysis

- **Blade:** Usually stainless steel (for strength, sterilizability).
- **Handle:** Plastic or metal; sometimes rubberized for grip.
- **Light source:** LED (cool, energy-efficient) or halogen.
- **Video parts** (if applicable): PCB, CMOS sensor, plastic lens housing.

b) Power Requirements

- Traditional: **AA batteries** inside handle.
- Video models: Rechargeable lithium-ion batteries.
- Use a **MultiMate** to check voltage levels and current consumption.

c) Ergonomics and Blade Geometry

- **Blade types:** Macintosh (curved) vs Miller (straight).
- Consider weight balance, grip design, switch placement.
- Evaluate ease of use for both right-handed and left-handed users.

5. System & Subsystem Redrawing

Objective: Visualize the product's structure and function via diagrams.

Steps:

- Create **block diagrams** showing:
 - Power flow (Battery → Switch → Light/Camera).
 - Physical structure (Handle ↔ Blade ↔ Light).
 - Data flow (in case of video laryngoscope: Camera → Display).
- Show **modularity**: how blade detaches, how electronics fit, etc.

6. 3D Modelling (CAD)

Objective: To digitally reconstruct the product using CAD software.

Steps:

- Use software **Solid Works**
- Model individual components:
 - Handle with battery compartment.
 - Blade with curvature and attachment features.
 - Optical components (lens or light housing).
- Combine into an **assembly model**.
- Create **exploded views** to show part relationships.

7. Testing

Objective: To understand limitations and operating conditions.

Tests to perform:

- **Battery life** and light intensity.
- **Stress testing** the blade (within safety limits).
- Test **joint strength** (blade-handle lock).
- Simulate clinical usage: visibility under different conditions (e.g., low light, fog).
- For video laryngoscope: check **image latency**, resolution, and lag.

8. Redesign Recommendations

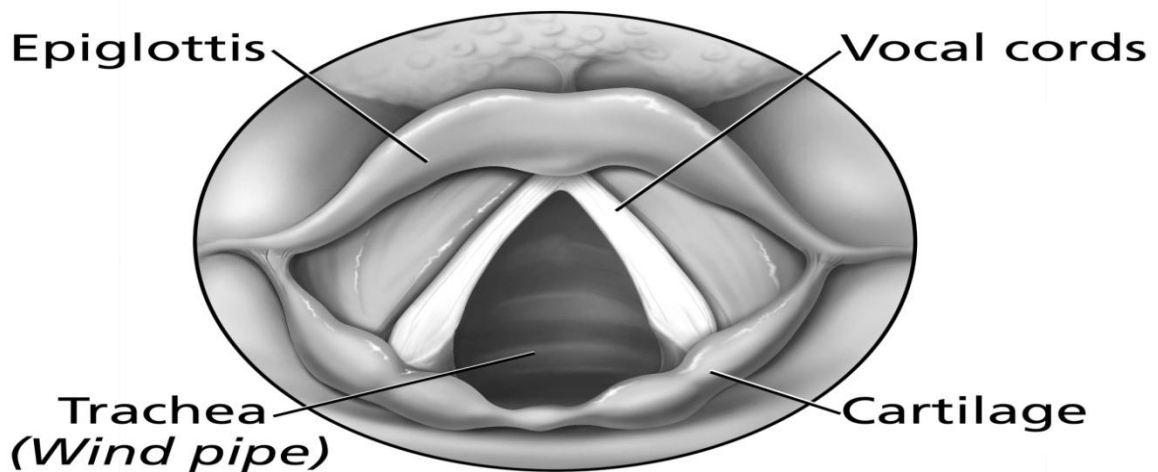
Objective: Identify areas of improvement based on reverse engineering.

Examples:

- **Material upgrade:** Use titanium blade for lighter weight.
- **Lighting:** Replace halogen with **longer-lasting LED**.
- **Anti-fog lens coating** or built-in **lens heater**.
- **Ergonomics:** Add **textured grip** or **angled handle** for better control.
- **Video enhancement:** Upgrade camera resolution or add wireless video transmission.
- **Sterilization ease:** Design for **modular disassembly** and autoclaving.

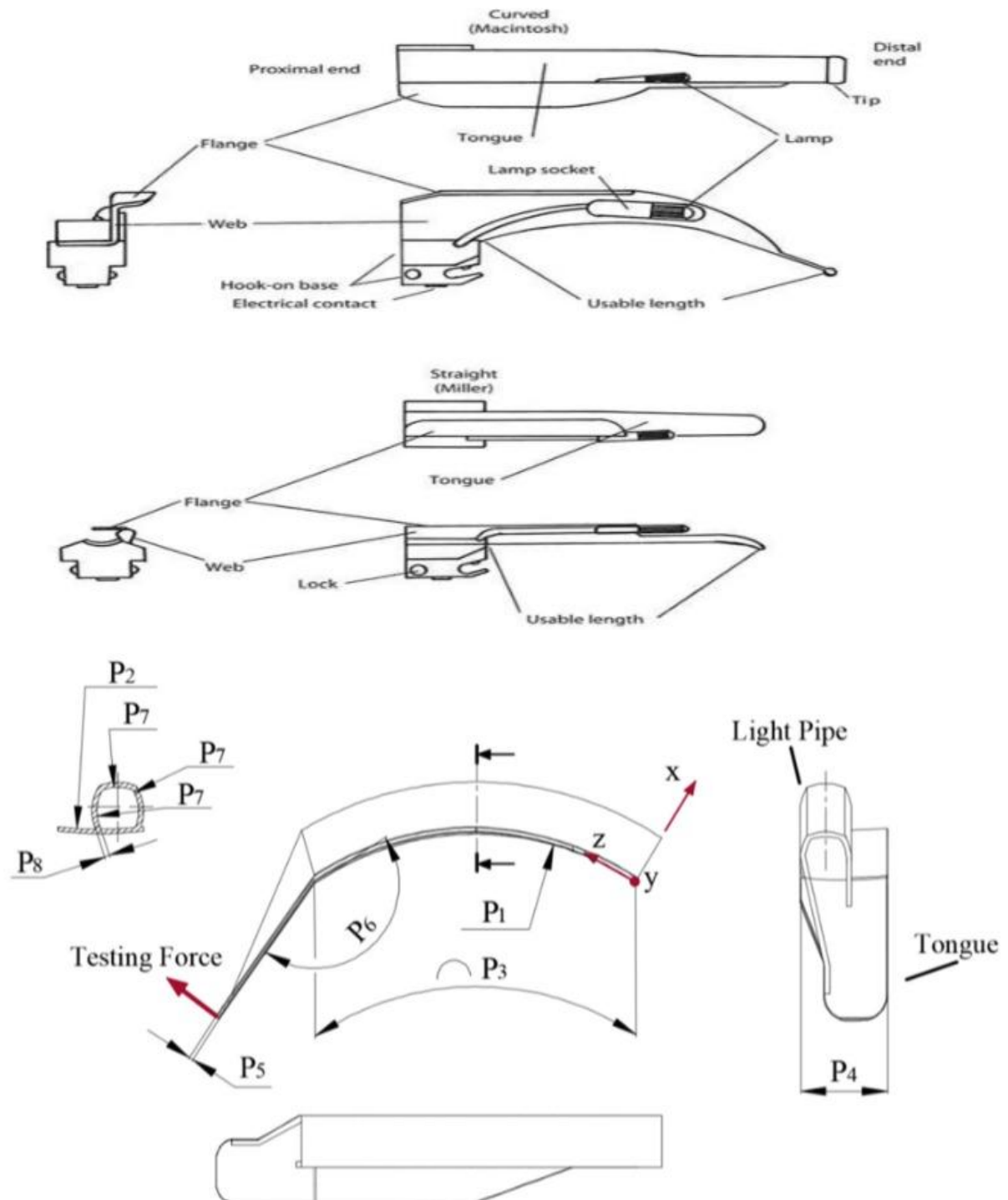
Table 1. Sets of parameter values for three blades

	Tongue					Light pipe		
Ref. plane	Curvature		Length	Width	Thickness	Inclination	Curvature	Thickness
	XZ P1 [mm]	XY P2 [mm]	XZ P3 [mm]	XY P4 [mm]	XZ P5 [mm]	XZ P6 [mm]	XY P7 [mm]	XY P8 [mm]
Adult blade	69.2	180	77.3	21	2	125	12.5	1.6
Pediatric blade	500	13	77.3	22	1.5	176	6	1.5
Customized blade	150	17	77.3	21	2.5	160	7	1.5



National Cancer Institute

MACINTOSH(CURVED)



SKETCH OF BLADE

SUBSYSTEM LEVEL

The Handle

The handle is the grip of the laryngoscope and contains the power source for the light.

- **Grip:** The handle is designed with ridges or a textured surface to provide a secure, non-slip grip during procedures.
- **Power source:** Typically, the handle houses batteries that provide electricity to the light source. A metal contact point on the handle connects to the blade to complete the circuit.
- **Size variation:** Handles come in various sizes, including short or "stubby" handles for patients with large chests and smaller "pencil" handles for pediatric patients.

The Blade

The blade is the part that is inserted into the patient's mouth to manipulate soft tissues and expose the larynx.

- **Base (Heel):** This is the end of the blade that attaches to the handle via a hinge-style connector.
- **Tongue (Spatula):** This is the main shaft of the blade that is used to depress the tongue and other soft tissues.
- **Flange:** A curved or angled part that extends from the side of the tongue. It is designed to guide instruments and keep the tongue out of the line of sight.
- **Tip:** The distal end of the blade, which is used to elevate the epiglottis, either directly or indirectly, to expose the vocal cords and glottis.

Blade types

There are two primary types of laryngoscope blades, which differ in shape and technique.

- **Macintosh (Curved) Blade:** The most popular type for adult intubation. Its curved shape is designed to be placed in the *vallecula* (the space between the tongue and the epiglottis). Lifting the handle pulls on a

ligament in this space, raising the epiglottis indirectly to reveal the vocal cords.

The light source

Illumination is crucial for clear visualization during a procedure.

- **Bulb-based:** In a conventional laryngoscope, a small incandescent bulb is located on the distal end of the blade. The electrical circuit is completed when the blade is attached and extended.
- **Fiber-optic:** Many modern laryngoscopes use a fiber-optic bundle to transmit light from a high-intensity bulb housed in the handle to the tip of the blade. This provides brighter, shadow-free illumination.

Hinge/Locking Mechanism

- **Function:**
 - Connects the blade to the handle securely.
 - Activates the light when the blade is opened into working position.
- **Type:** Snap-fit or screw-lock



Limitation and challenges

- First time using CAD software
- Geometry overlapping
- Troubleshooting errors
- Every printer must have its post processing tool kit
- Designing smooth and realistic hinge mechanism was a challenge in CAD
- Scaling down the model led to reduced size of holes
- Filament got stuck
- PLA/ABS/PETG are not medical-grade cannot be used directly for patient application

Redesign

1. Make it Easier to Use

- Lighter materials like carbon fiber
- A handle that fits better in the hand
- Non-slip grip for steady control

2. Improve What You Can See

- Built-in **HD camera** with a wide-angle lens
- Anti-fog, clear-view blade coating
- Real-time screen or wireless video to help guide intubation

3. Brighten It Up

- Adjustable LED lighting
- Optional special lighting modes (e.g. UV or infrared for surgery)

4. Keep It Clean

- Blades that can be removed and sterilized
- Disposable blade options for emergencies